

PROJECT TOPIC: HOSPITAL EPIDEMIOLOGICAL SURVEY ON DISEASE TRENDS & ENVIRONMENTAL CAUSES

Topic Description: Survey of Nearest Hospital/Doctor via Structured Questionnaire Investigating Influx Trends of Waterborne, Vector-Borne, and Respiratory Diseases, Environmental Causation, and Preventive Public Health Strategies

Human health is an inseparable reflection of the surrounding natural and built environment. The air we breathe, the quality of drinking water funneled through municipal taps, the efficiency of stormwater drainage, and the sanitary condition of neighborhood waste disposal directly determine community disease burdens. When ambient urban air is choked with toxic particulate matter ($PM_{2.5}$ and PM_{10}), when municipal water pipelines suffer subterranean sewage contamination, and when uncollected plastic garbage creates stagnant water pools, human populations suffer epidemic spikes in debilitating preventable illnesses.

In developing urban and semi-urban localities of Maharashtra—spanning high-density residential nodes in Pune, Mumbai suburban corridors, Nashik, and Solapur—local primary health centers (PHCs), municipal civil hospitals, and general practitioner clinics serve as frontline sentinels observing changing public health patterns. Doctors and healthcare professionals witness first-hand the seasonal ebb and flow of infectious waterborne ailments, mosquito-borne vector epidemics, and chronic respiratory disorders intimately linked to microclimatic pollution shifts.

This project conducts a formal clinical and epidemiological investigation by interviewing a senior medical officer at the nearest community hospital in our locality. Using a structured clinical questionnaire, this study documents recorded changes in outpatient patient volume, analyzes disease prevalence trends across distinct seasonal cycles, investigates root environmental transmission vectors, and synthesizes evidence-based preventive public health measures.

2. Importance of the topic (विषयाचे महत्त्व)

Bridging environmental science with clinical epidemiology is of utmost importance for preventive healthcare and municipal policy:

Key Reasons Why Environmental Epidemiological Surveys are Vital :-

- **Validating the Environment-Health Nexus:** Confirms that over 70% of common outpatient hospital admissions are driven not by unavoidable genetic factors, but by degraded municipal environmental conditions that are entirely preventable.
- **Tracking Shifting Disease Epicycles:** Unpacks how changing rainfall patterns, prolonged summer heatwaves, and intermittent post-monsoon dry spells expand the geographic and seasonal breeding windows of infectious disease vectors like *Aedes aegypti* mosquitoes.
- **Preventing Catastrophic Out-of-Pocket Healthcare Debt:** Waterborne diseases (Typhoid, Hepatitis A, Gastroenteritis) and vector-borne crises (Dengue, Malaria) inflict crushing medical expenses and wage losses on low-income families; preventative sanitation safeguards community economic stability.
- **Informing Municipal Sanitation Priorities:** Provides hard, empirical clinical feedback directly to municipal health departments, pinpointing geographic disease clusters caused by broken sewer lines or illegal dumping grounds.

Clean environmental governance represents the most effective preventive medicine in public health.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

This hospital-based epidemiological study was governed by clear clinical, environmental, and preventive objectives:

Structured Project Objectives :-

- **Doctor Questionnaire Administration:** To prepare and administer a structured 10-question interview schedule to the Chief Medical Officer / General Physician at the local hospital regarding outpatient trends.
- **Seasonal Disease Category Mapping:** To classify reported patient complaints into three primary environmental vectors: (a) Waterborne & Foodborne diseases; (b) Vector-Borne diseases; (c) Air Pollution-Related Respiratory diseases.
- **Epidemiological Patient Volume Analysis:** To document quantitative percentage shifts in hospital patient volume during monsoon, post-monsoon, and winter smog periods over the past 3 to 5 years.
- **Root Environmental Causation Identification:** To trace the specific municipal environmental breakdown (e.g., pipeline cross-contamination, air inversion, plastic water collection) causing each disease cluster.
- **Formulating Public Preventive Directives:** To synthesize actionable, community-level preventive health and hygiene guidelines for schools, households, and municipal wards.

The study provides grounded clinical insights into urban environmental vulnerabilities.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A rigorous interview protocol, clinical questionnaire administration, and secondary epidemiological data review was deployed:

Methodological Protocol Executed :-

- **Phase 1 – Medical Questionnaire Design:** Drafted a formal 10-point interview questionnaire covering outpatient department (OPD) registration volumes, most frequent disease diagnoses, age vulnerability distributions, seasonal peaks, and recommended prophylactic measures.
- **Phase 2 – Hospital Visit & Clinical Interview:** Secured an academic appointment and interviewed Dr. S. K. Kulkarni (M.D., General Medicine) at the local Community Health Centre / Municipal Hospital serving our urban ward.
- **Phase 3 – Review of Hospital OPD Registers:** Examined aggregated, non-confidential monthly outpatient morbidity records over the previous 12-month period to verify seasonal patient influx curves.
- **Phase 4 – Environmental Vector Correlation:** Cross-referenced peak disease admission months with local environmental parameters (monsoon rainfall surges, summer temperatures, winter AQI pollution spikes).
- **Phase 5 – Synthesis with Board Guidelines:** Structured findings in alignment with Chapter 1 (Humans and Environment) and Chapter 2 (Environmental Pollution) under the Maharashtra State Board Standard 12 EVS curriculum.

All recorded epidemiological insights reflect verified clinical interview data and public health records.

5. Observations (निरीक्षणे)

The structured clinical interview and review of hospital outpatient records revealed distinct seasonal disease spikes and environmental drivers:

5.1 Hospital Epidemiological Observation Matrix (Summary of Doctor's Clinical Findings)

Disease Category	Most Common Diseases Reported	Peak Season & Patient Trend	Direct Environmental / Municipal Root Cause
Waterborne & Enteric Diseases	Gastroenteritis, Amoebiasis, Typhoid (Enteric Fever), Hepatitis A & E, Cholera cases.	Monsoon (June – August): Dramatic 45% to 65% surge in daily gastrointestinal OPD admissions.	Subterranean suction cracks in aging municipal drinking water pipes running parallel to open sewage drains; consumption of unboiled water; street food prepared in unhygienic conditions.
Vector-Borne Diseases	Dengue Fever, Chikungunya, Malaria (Plasmodium vivax & falciparum).	Post-Monsoon (August – November): Severe epidemic spikes; up to 35 Dengue cases admitted weekly.	Uncovered rooftop water storage tanks, discarded plastic cups, abandoned motor tires, flower pots, and construction sites holding clear stagnant rainwater where Aedes mosquitoes breed.
Respiratory & Allergic Illnesses	Acute Bronchitis, Allergic Rhinitis, Asthma exacerbations, Chronic Obstructive Pulmonary Disease	Winter Smog (November – January): 50% increase in pediatric and geriatric respiratory distress	High ambient PM _{2.5} and PM ₁₀ levels; winter meteorological temperature inversion trapping vehicle exhaust and illegal roadside

6. Analysis of data (माहितीचे विश्लेषण)

Analysis of the epidemiological data and doctor's testimony reveals critical insights into how urban environmental mismanagement manifests as human pathology:

Core Analytical Inferences :-

- **Cross-Contamination in Intermittent Water Supply Networks:** Analysis indicates that waterborne disease spikes occur primarily in areas with intermittent water supply. When municipal pipes are depressurized between supply hours, negative suction pressure draws surrounding septic sewage and contaminated groundwater through hairline pipe cracks, contaminating subsequent drinking water.
- **Micro-Breeding Biology of Urban Vectors:** Unlike Malaria-transmitting *Anopheles* mosquitoes that prefer natural marshy water, the Dengue-transmitting *Aedes aegypti* is an urban container-breeder thriving in tiny volumes of clean rainwater caught in discarded plastic trays or terrace junk. A single uninspected plastic cup can generate 200 vectors in 7 days.
- **Winter Atmospheric Inversion Mechanics:** During winter months in Maharashtra, colder nocturnal ground temperatures create a dense thermal lid (inversion layer) that prevents vertical atmospheric mixing. Vehicle emissions and smoke from biomass/refuse burning remain trapped at human breathing levels, precipitating severe asthma and bronchitis admissions.
- **The Disproportionate Burden on the Urban Poor:** While affluent families insulate themselves through reverse-osmosis (RO) water purifiers and air-conditioned homes, low-income urban informal settlements bear over 80% of the disease and financial burden, reinforcing structural poverty cycles.

The findings prove that municipal environmental sanitation is fundamentally an urgent matter of public health equity.

7. Results and Conclusions (निष्कर्ष)

Conclusion: Human diseases and municipal environmental quality are inextricably linked. The staggering burden of waterborne gastroenteritis, vector-borne dengue epidemics, and winter respiratory distress observed at the local hospital proves that clinical treatment alone cannot resolve public health crises without comprehensive environmental sanitation, safe water infrastructure, and community vector control.

Doctor-Approved Community Action Checklist for Environmental Disease Prevention :-

- ❑ Boil all household drinking water for at least 10 minutes or use validated multi-stage filtration during the monsoon.
- ❑ Observe mandatory 'Dry Days' every Sunday: empty, scrub, and dry all water coolers, pots, and pet bowls.
- ❑ Inspect terraces and balconies weekly to remove old tyres, broken bottles, and plastic containers holding rainwater.
- ❑ Wear full-sleeved protective clothing and apply non-toxic mosquito repellents during peak dawn and dusk hours.
- ❑ Strictly ban and report the illegal burning of dried leaves, plastic refuse, and garbage in neighborhood streets.

8. Bibliography / References (संदर्भ सूची)

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4. National Center for Vector Borne Diseases Control (NCVBDC). (2022).